CAS (Heimdall) – system functional description

Inlet system: water intake and pen level control

ScaleAQ Automation Document CAS-FD-10, rev. A

October 21, 2026

Abstract

This document describes how the inlet system of the CAS (Heimdall) closed fish pen works and how the control system controls it. Three Framo inlet pumps fill the pen with sea water; the Heimdall PLC keeps the water level inside the pen a few centimetres above the sea outside (the *over-height*) by setting one common speed for the pumps. The document lists every instrument and actuator, the control principle, the start and stop sequences, the pump state machine, the interlocks, the alarms and how faults are handled. Where the sources disagree, or a value is not yet decided, the text marks an open issue (**OIn**) that is listed in Section 6.

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1 Introduction

1.1 Purpose and readers

The document is the system functional specification of the inlet system in the sense of [COO] §14.3: a description of the physical system (*block diagram description*), of what the control system does with it (*functional diagram description*) and of the functional and non-functional requirements. It is written for

- **engineers and the project team**, who implement, test and commission the PLC program;
- **operators**, who need to know what the system does on its own and what they must do;
- **suppliers** (Framo for the inlet pumps), who must agree the interface and the behaviour at the boundary.

The Norwegian functional description [FD] is authoritative on the *intent*. This document describes the intent in English, shows what the PLC program [PLC] implements today (branch in review, 2026-10-21) and lists where the two differ.

1.2 Scope

In scope: the water intake from the three intakes to the pen (Framo inlet pumps PM-31/32/33, their local controllers and vacuum systems as seen from the PLC), the level measurement inside and outside the pen, the level controller LIC01, the start and stop of the pen and of each pump, the communication with the Framo controllers and the related alarms.

Out of scope (separate FDs): the oxygen system (side stream, oxy pumps PU41–43, OIC01), the outlet dampers and their hydraulics, the bilge system in the inlet boxes, and the power distribution. They appear here only where they interface with the inlet system: the oxygenation permit per line, the level setpoint per operating mode from the outlet, and interlocks from the inlet boxes.

1.3 Sources

Table 1: Sources. The IDs are used in the text.

ID	Document	Revision / date
[FD]	P-15169-CAS-FD-01 *Funksjonsbeskrivelse CAS*, §3.2.1 Vanninntak, §3.2.2 level control, tables 5–10 (Norwegian)	rev. B, 13.05.2026
[AL]	P-15169-CAS-FD-02 *Alarmliste CAS Heimdall*	rev. A
[FRAMO]	Framo, *Spesifikasjon: kontrollsystem (prototype)* 0000_CAS_SPEC_CTRLSYS, signal list and state table, with ScaleAQ review comments	rev. B, 08.10.2025
[FIL]	Framo interface list (early signal list)	01.09.2025
[PID]	CAS P&ID and loop diagrams: CAS-PID-100 inlet, -400 pen, -600 utilities, CAS-LAY-700 network, CAS-LD-801 loop LT01, CAS-PID-950 open points	rev. A, 2026-10-16
[DES]	CAS design basis (System Modelling), generated from design_basis.toml	v2.3.12
[PDS]	CAS program design (dataflow of the level loop, start command path)	2026-10-15
[PLC]	Heimdall PLC program (CODESYS, application PFC200_Sim): FB.Inlet.InletControl, InletSupervisor, InletSensors, FB.Pen.PenControl, PRG.FramoDataProcessing, GVL ConfigValues	branch in review, 2026-10-21
[EL]	Electrical drawings 15169-792.001 (PLC/RIO1 cabinet), 792.002 (RIO2/3 cabinet), 874.001 (Framo pump cabinet)	792.001 rev. 1, 792.002 rev. 0
[COO]	J. Cooling, *Software Engineering for Real-Time Systems*, 2020, §14.3 System functional specifications	2020

1.4 Abbreviations and terms

Term	Meaning
CAS	Closed Aquaculture System; Heimdall is the control system of the CAS pen
Over-height, ΔH	Water level inside the pen minus sea level outside [m or cm]
ΔH_i, pair i	Over-height measured at inlet i: LT0i − LT2i
PV, SP	Process value, setpoint
LIC01	Level indicating controller of the pen (PI)
Framo PCCn	Pump control cabinet of inlet n: Beckhoff CX8200 PLC and ABB ACS880 drive
PFC200	WAGO 750-8212, the Heimdall PLC
RIO	Remote I/O: WAGO 750-362 Modbus TCP couplers (CommonRIO, InletRIO1–3)
OPC UA	Communication between the PLC and the Framo controllers (and the HMI)
HMI	Operator interface (Heimdall HMI / SiteControl)
VSD	Variable speed drive
REMOTE, HOLD, ...	Pump states published by the Framo controller (Section 3.4)
O-35, O-45, O-60, S-75	Operating modes: operation (O) or service (S), with residence time in minutes
TBD	To be decided; see the open issues

1.5 Revision note

Rev.	Date	Change
A	2026-10-21	First issue. Based on [FD] rev. B, [FRAMO] rev. B, [PID] rev. A and the PLC branch in review (heartbeat, HOLD, per-pair PV and SafeOutput changes of 2026-10-15). Includes the review and simulator test of 2026-10-21.

2 Block diagram description

2.1 The process in brief

The pen PEN01 is a closed bag of about 20 000 m^3 (Ø 50.5 m, depth 17.3 m) with a Ø 5.5 m outlet in the bottom. Sea water is pumped in from three separate intakes at 10–30 m depth. Each intake has a fish screen (XF-3n), a tunnel to a Framo inlet pump (PM-3n, rated 20 000 m^3/h at 130 rpm, 60 kW rim motor) and a riser with a gooseneck over the floating collar into the pen.

The water inside the pen is kept a few centimetres higher than the sea outside. This *over-height* presses the bag outwards so that it keeps its shape, and it drives the outflow through the bottom outlet. The outlet dampers stand at a fixed opening for each operating mode; the level is therefore controlled with the *pump speed*: more speed gives more inflow, a higher level and more outflow, until inflow and outflow balance [FD] §3.1.

Before a pump can lift water over the collar, the air in the riser must be removed. Each inlet has a vacuum (air evacuation) system run by the Framo controller: a vacuum chamber TV-3n on top of the riser, an ejector EJ-3n driven by compressed air through valve XV3nA, level switches LSL3n/LSH3n and a vent valve XV3nB that breaks the vacuum when the pump stops, so the pen does not drain back through the intake [FD] §3.2.1.1.

The operating modes set the flow through the pen and the level setpoint (Table 2).

Table 2: Operating modes [FD] Tabell 10, with the operating point of the design basis [DES].

Mode	Res. time [min]	Flow [m³/h]	SP ΔH [cm]	Pumps [–]	Speed n_0 [rpm]	Damper [%]	Comment
O-35	35	34 000	6.0	3	73.7	100	flow above the [DES] maximum[OI12]
O-45	45	27 000	2.5	2	87.8	100	
O-60	60	20 000	0.5	2	65.0	91	5 mm setpoint[OI7]
S-75	75	16 000	0.5	2	52.0	73	service mode[OI7]

With two pumps running, [FD] asks for the pumps to be alternated so that the third does not stand still for too long (HOLD in [FD])[OI15].

2.2 Physical units and networks

Figure 1 shows the physical units and how they are connected; Table 3 gives the role of each.

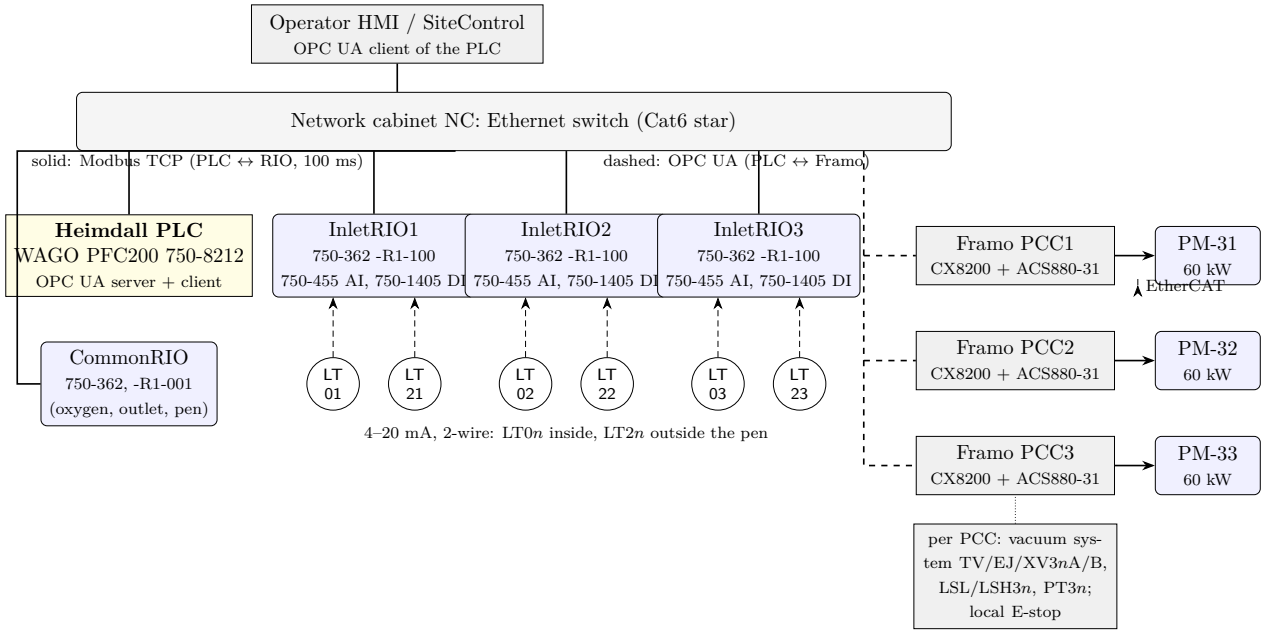


Figure 1: Physical units and networks of the inlet system (after [PID] CAS-LAY-700 and [EL]). IP addresses are not on any drawing[OI23].

2.3 Simplified process and instrument diagram

Figure 2 is a simplified P&ID for one inlet ($n = 1$); inlets 2 and 3 are identical. Tags follow ISA-5.1 as on the P&ID set [PID]. Bubbles: plain = field instrument, with a bar = function in the Heimdall PLC, grey = function in the Framo controller.

Table 3: Physical units of the inlet system.

Unit	Role
Framo PCC1–3	One cabinet per inlet [EL] 874.001: Beckhoff CX8200 PLC (TwinCAT 3, OPC UA server) and ABB ACS880-31-105A-3 drive for the rim motor of PM-3n, local emergency stop pushbutton with Safe Torque Off (XSTO). Runs the pump's own start and stop sequences, the vacuum system and the protections, and publishes the pump state. Framo scope.[OI19]
Framo inlet pumps PM-31/32/33	SR2000 rim-driven propeller pumps, 40–130 rpm, 20 000 m^3/h at 130 rpm, 72-pole motor (78 Hz at 130 rpm) [DES].
Vacuum system (per inlet)	TV-3n, EJ-3n, XV3nA (air to ejector), XV3nB (vent), LSL3n, LSH3n, PT3n (compressed air). Controlled by the PCC; the PLC only reads status.[OI11]
Heimdall PLC	WAGO PFC200 750-8212 in the PLC/RIO1 cabinet (UPS backed). Runs the level loop, the pen sequence and the pump supervision. No local I/O.
InletRIO1–3	750-362 Modbus TCP couplers, one per inlet. 750-455 analog inputs for LT0n (word 0) and LT2n (word 1); 750-1405 for cabinet power status (word 4) and bilge switches (word 5). Polled by the PLC every 100 ms.
CommonRIO	750-362 coupler for the common signals (oxygen, outlet, analysis). Not used by the inlet functions.
Level transmitters	LT01–03 inside the pen (VEGAPULS radar on the gooseneck), LT21–23 outside (submersible pressure probe), one pair per inlet, 0–2 m, 4–20 mA 2-wire.[OI18]
HMI / SiteControl	Operator interface; reads and writes the PLC's variables over OPC UA.

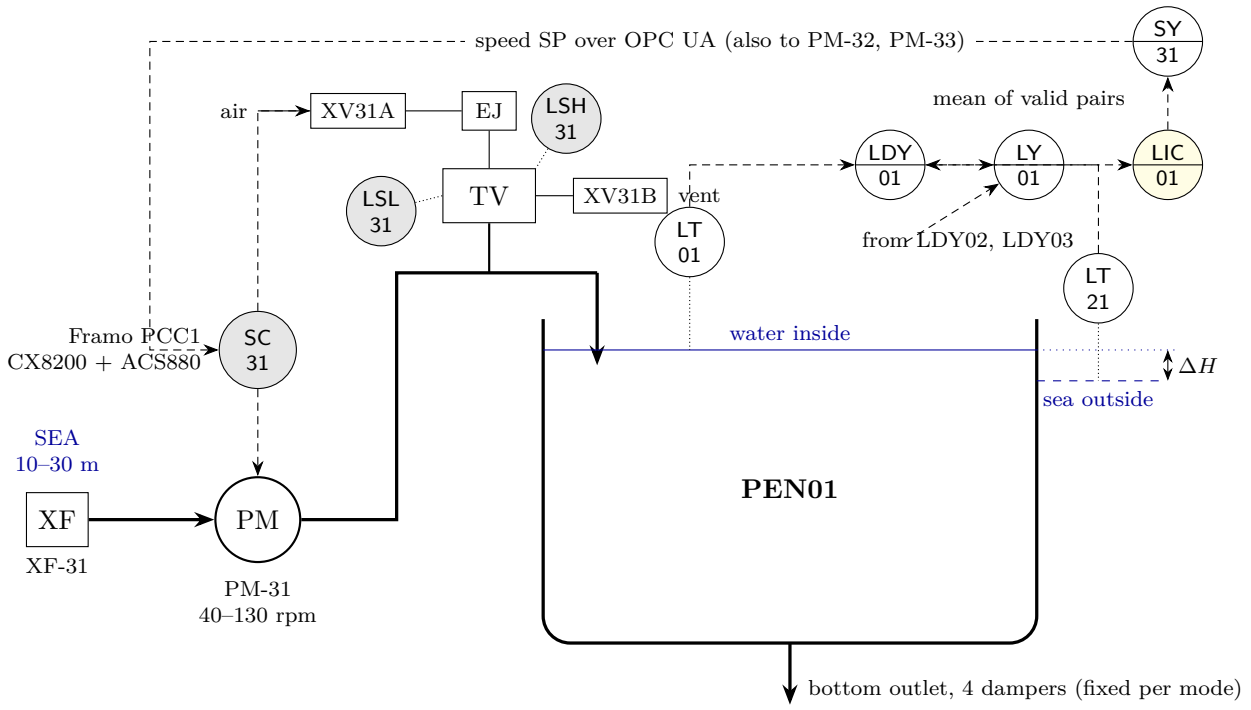


Figure 2: Simplified process and instrument diagram, inlet 1 ([PID] CAS-PID-100 and -400). LDY0n forms the over-height of pair n, LY01 the mean of the valid pairs, LIC01 the common speed setpoint SY3n of all pumps in level control.

3 Functional diagram description

3.1 Top-level functions

Figure 3 shows the functions of the inlet system and the data between them. The functions F1–F3 run once per pen; F4 and F5 once per pump. The Framo controller (grey) is outside the Heimdall PLC.

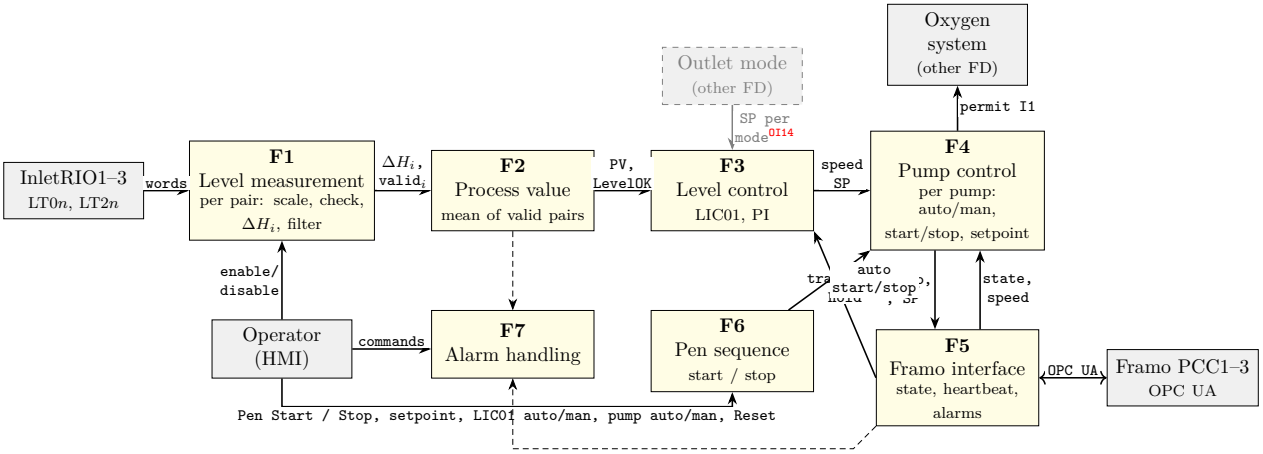


Figure 3: Top-level functional block diagram of the inlet system [COO] Fig. 7.8 style. F7 receives alarm conditions from F1–F5 (only two drawn).

Table 4: Functions and where they are implemented in [PLC].

	Function	What it does	PLC code unit
F1	Level measurement	Scales LT0i and LT2i, checks their health, forms ΔH_i and filters it.	`FB.Inlet.InletSensors`, `AnalogSensor`
F2	Process value	Mean of the valid ΔH_i; LevelControlOK.	`FB.Pen.PenControl`
F3	Level control	LIC01 with tracking, hold, resume and safe output.	`InletSupervisor.LevelControl`
F4	Pump control	Auto/manual, start/stop latch, speed setpoint per state, start/stop timeouts, oxygen permit.	`InletControl.InletPumpControl`
F5	Framo interface	Writes commands, reads state, alarms and values; heartbeat supervision.	`PRG.FramoDataProcessing`, `InletControl.Heartbeat`
F6	Pen sequence	Pen Start/Stop latch, auto start/stop of the pumps in auto.	`InletSupervisor`
F7	Alarm handling	Alarm state and acknowledgement per alarm.	`AlarmManagement` instances

3.2 Level control loop LIC01

Principle. One PI controller, LIC01, holds the over-height at the setpoint. Its output is *one* speed setpoint in rpm, sent to every pump that is in automatic and in REMOTE; all those pumps therefore run at the same speed [FD] §3.2.2. Figure 4 is the control scheme.

Process value. At each inlet i the PLC computes the over-height of the pair from the scaled,

unfiltered readings and filters it with a 2nd-order Butterworth low-pass (cut-off $f_c = 0.05$ Hz):

$$\Delta H_i = \mathrm{LT0}i - \mathrm{LT2}i, \qquad \mathrm{PV} = \Delta H_{\mathrm{avg}} = \frac{1}{|V|} \sum_{i \in V} F\{\Delta H_i\}$$

where V is the set of *valid* pairs. A pair is valid when both transmitters are enabled and healthy (750-455 status bits 1..0 = 00) and the pair itself is enabled. If either transmitter of a pair is lost, the whole pair drops out of the mean[DEC1]. [FD] asks to filter each transmitter before the difference; since the filter is linear, filtering the difference gives the same result while both are healthy, and it restarts cleanly when a pair returns [DES] §7.

Setpoint. The over-height setpoint in m (`Pen.Command.LevelSetpoint`, retained, default 0.06 m = O-35). [FD] wants it selected from the operating mode, with a manual override; today the operator enters it[OI14].

Output. Speed setpoint 40–130 rpm (the pump's speed range). While LIC01 is in manual its output is taken, in this order of priority, from: (1) the operator's manual output; (2) `SafeOutput` (60 rpm) when no valid pair is left[DEC3]; (3) the output frozen when the heartbeat to a pump in level control was lost; (4) the mean actual speed of the pumps in level control for one cycle after the freeze (resume); (5) the mean actual speed of the starting pumps (tracking). Manual→auto is bumpless.

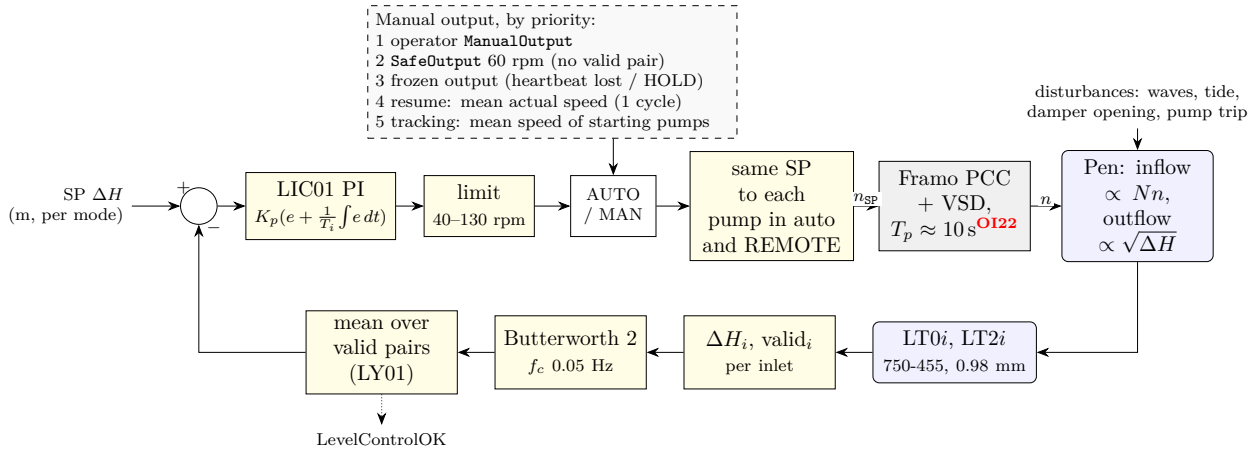


Figure 4: Control scheme of LIC01 ([PDS] SMs 1–10, [DES] §7). N = number of pumps in level control.

3.3 Pen start and stop sequence

Figure 5 shows the sequence that the operator's Pen Start and Pen Stop commands run. Pumps in manual are left out of it; the operator starts and stops them one by one.

3.4 Framo pump state machine

The Framo controller owns the pump state and publishes it as one Boolean per state; the PLC turns the flags into `Inlet[i].InletPump.State` and asks for transitions with Start, Stop and Reset. Figure 6 is the state diagram of [FRAMO] rev. B; red dashed labels mark where [FD] or the PLC expect something else. Table 5 gives what the PLC does in each state.

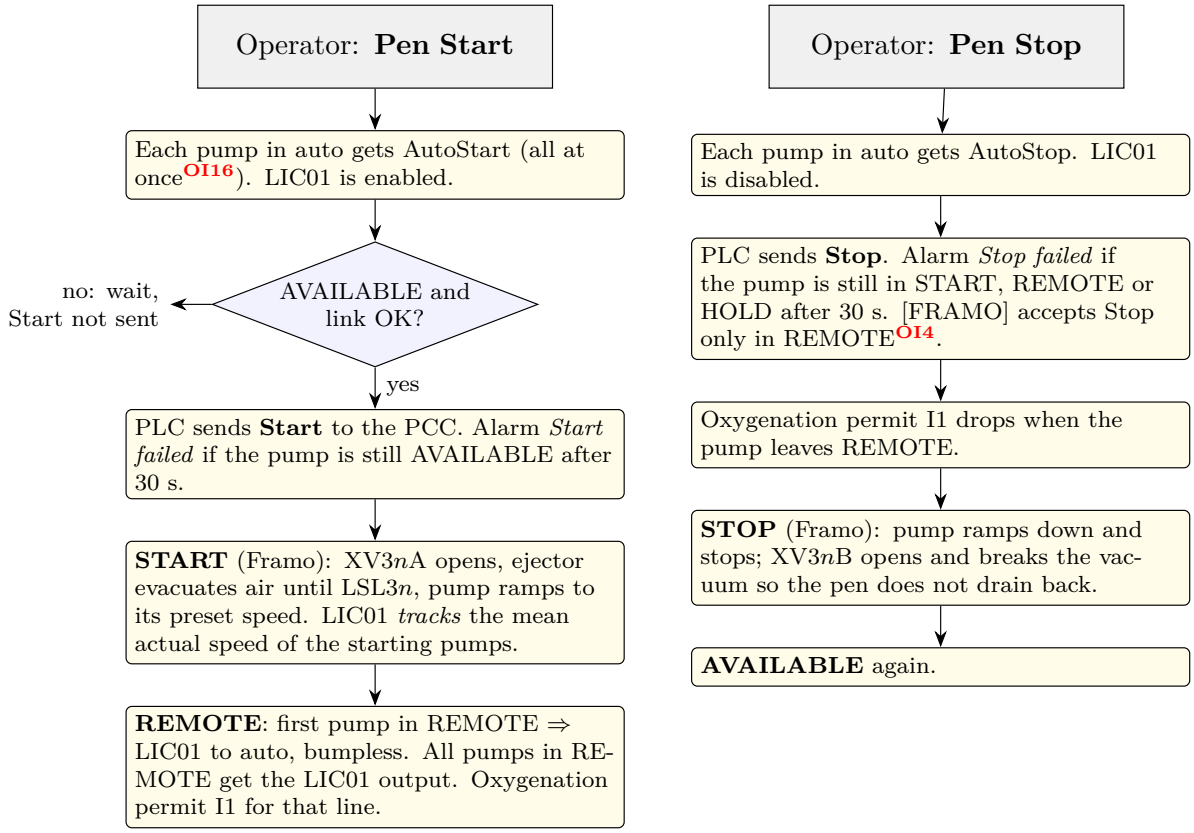


Figure 5: Pen start (left) and stop (right) sequence [FD] §3.2.1, [PLC] InletSupervisor/InletControl.

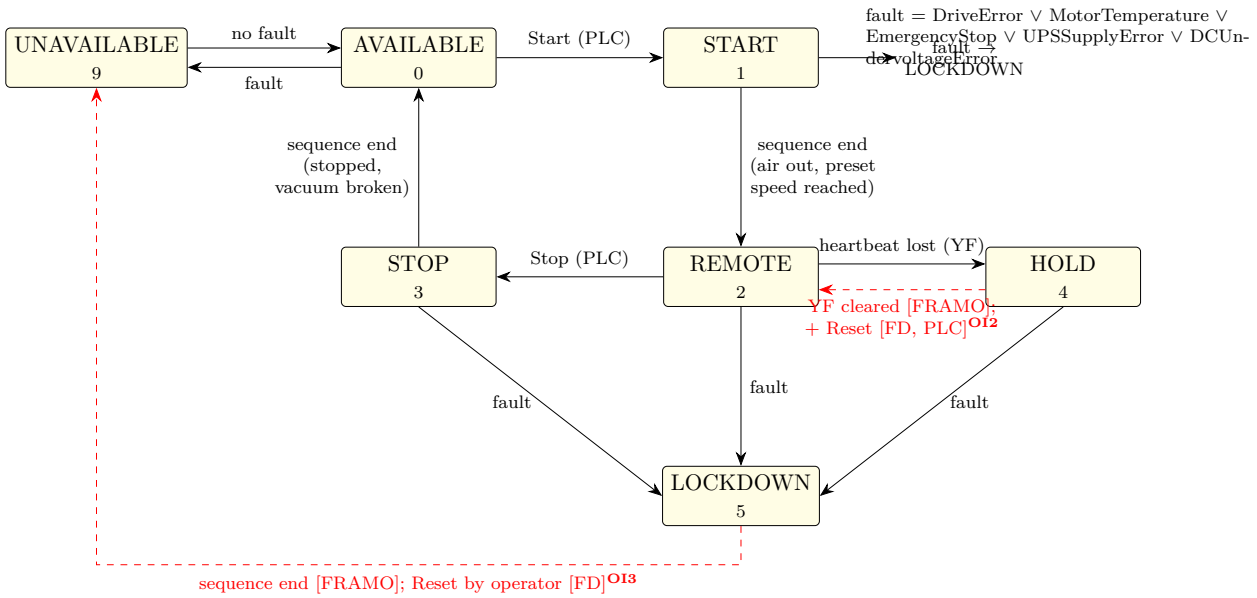


Figure 6: State transition diagram of the Framo inlet pump [FRAMO] rev. B p. 6, [COO] §7.2.3. Numbers are the values of `ENUM.InletPumpState`. In addition the PLC shows a pump UNAVAILABLE whenever it has no heartbeat answer from that PCC, whatever the PCC publishes.

Table 5: PLC action per pump state ([PLC] `InletControl.InletPumpControl`).

State	Speed setpoint sent	Start allowed	In LIC01	Notes
UNAVAILABLE	0	no	no	Also shown while the link (heartbeat) is down.
AVAILABLE	actual speed	yes	no	Setpoint follows the actual speed so the take-over is bumpless [FD] Tabell 5.
START	actual speed	yes	yes	LIC01 tracks the mean speed of the starting pumps.
REMOTE	LIC01 output (auto) or operator setpoint (manual)	yes	yes	The only state in which the PLC sets the speed. Oxygenation permit I1 with a live link.
HOLD	actual speed	yes	no	PCC runs its preset safe speed; LIC01 frozen; oxygenation stops.
STOP	0	no	yes	
LOCKDOWN	0	no	no	Needs Reset after the fault is cleared[OI3].

3.5 Heartbeat

The PLC and each PCC supervise the link between them with a heartbeat. As implemented in the PLC (2026-10-15), the heartbeat is *one shared Boolean* `Framo_n.HeartbeatToggle` (Figure 7):

- the PLC sets it when it has been FALSE for `HeartbeatInterval` (1 s);
- the PCC clears it when it is TRUE;
- the bit always says whose turn it is, so after a (re)connect neither side waits for the other;
- only the PCC's clear (falling edge) proves that the PCC is alive. When the PCC has not cleared the bit for `HeartbeatTimeout` (10 s), the PLC raises *Communication loss*, shows the pump UNAVAILABLE, sends no Start, freezes LIC01 if the pump was in level control and removes the oxygenation permit of the line;
- the PCC, seeing no set from the PLC within its timeout, moves the pump to HOLD and runs it at a preset safe speed.

The timeout is 10 s instead of the 5 s DUT default because the round trip through the CODESYS Data Sources takes about 4.6 s in the test rig. [FRAMO] rev. B lists `HeartbeatToggle` (BOOL, RW), `HeartbeatInterval` and `HeartbeatElapsed`; [FD] describes an INT counter 0–32767. The baseline must be agreed with Framo[OI1].

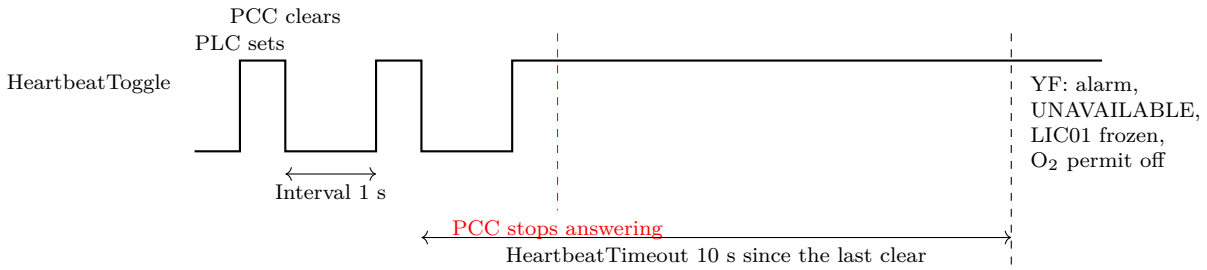


Figure 7: Heartbeat as implemented in [PLC]: one shared bit, the PLC sets it, the PCC clears it.

3.6 Operating modes

The operating modes (Table 2) are selected on the outlet function. A mode sets the damper opening and, per [FD], the level setpoint of LIC01 and the number of pumps (2 or 3). In [PLC] the outlet has the mode table with damper positions, but neither the level setpoint nor the number of pumps is taken from the mode yet[OI14][OI15].

4 Functional requirements

4.1 Processes

P1 The system shall pump sea water into the pen with up to three Framo inlet pumps; normally 2 or 3 run, depending on the operating mode.

P2 The system shall hold the over-height ΔH_{avg} at the setpoint by setting one common speed for all pumps in level control (LIC01).

P3 The system shall start and stop the pumps in automatic from one Pen Start / Pen Stop command, and let the operator start, stop and set the speed of a pump in manual.

P4 The system shall supervise the link to each Framo controller and react to its loss (HOLD).

P5 The system shall give each oxygen line its permit only while its inlet pump runs in REMOTE with a live link (interlock I1).

P6 The system shall present the pump state, the Framo values and alarms, and the levels on the HMI.

4.2 Inputs

Table 6: Inputs to the inlet functions (details in Tables 12 and 13).

Input	Source	Used for
LT01–03, LT21–23	InletRIO n words 0, 1	PV of LIC01, display, alarms
Pump state flags (7 per pump)	PCCn, OPC UA	F4, tracking, hold, start permissive
MotorVelocity	PCCn, OPC UA	tracking, bumpless transfer, display
Framo alarms (11 per pump)	PCCn, OPC UA	alarms, info panel
HeartbeatToggle	PCCn, OPC UA	link supervision
Vacuum status (LSL, LSH, V001, V002, air pressure)	PCCn, OPC UA	display; overfill alarm[OI11]
Drive values (current, power, ...)	PCCn, OPC UA	info panel
24 V field breaker XF05 status	InletRIO n word 4.0	alarm (it feeds the LT loops)
Operator commands and parameters	HMI, OPC UA	Table 14

4.3 Outputs

- Per pump, to PCC*n* over OPC UA: Start, Stop, Reset, SpeedSetpoint [rpm], HeartbeatToggle (set).
- To the oxygen system: `Inlet[i].InletPump.Status.EnableOxyControl` (permit I1).
- To the HMI: states, PV, setpoint, LIC01 output and mode, alarms.

The inlet system has no hard-wired outputs; the PLC has no command path to the pumps other than OPC UA. A lost link therefore always leaves the pump to the PCC (HOLD).

4.4 Startup and shutdown

PLC start (power-up, download). Configuration and the level setpoint are retained. The pen start latch is not retained: after a PLC restart no pump is started automatically. [FD] Tabell 2–3 asks for auto-restart after a power failure, in a sequence to avoid simultaneous inrush currents; this is not implemented[OI16]. The PLC shows every pump UNAVAILABLE until its PCC has answered the heartbeat once, so no Start is sent on stale data.

Normal start and **normal stop**: Figure 5.

Pump start while others run. A pump started (in auto or manual) while another pump is already in REMOTE does not interrupt automatic control; it joins LIC01 when it reaches REMOTE.

Emergency stop. Each PCC has a local E-stop pushbutton acting on the drive's Safe Torque Off; [EL] 874.001 notes that multiple E-stop actuators are connected in series. Whether the E-stops stop all three lines, and how the oxygen line is stopped on E-stop through the Heimdall PLC, is to be confirmed[OI13]. The PCC reports `EmergencyStop`; the pump goes to LOCKDOWN (or UNAVAILABLE if it was not running).

4.5 Fault identification and recovery

Table 7: Faults, detection, automatic response and recovery.

Fault	Detection	Automatic response	Recovery
One level transmitter bad	750-455 status bits ≠ 00 (wire break, over-range) for 1 s	Quality alarm; its *pair* drops out of the PV; LIC01 continues on the remaining pairs	Automatic when healthy; the filter restarts at the current value (no dip)
Transmitter disabled by operator	Config.Enable FALSE	Pair drops out	Operator enables it
All pairs lost	no valid pair	Alarm *Level measurement lost* (LAL01); LIC01 to manual at `SafeOutput` 60 rpm (interlock I6)	Back to auto bumplessly when a pair is valid
InletRIO *n* link lost	Modbus TCP `CommOK` FALSE	Alarm *InletRIO comm. error*; level words keep the last value and the pair stays valid[OI17]	Automatic
24 V field breaker XF05 tripped	InletRIO word 4.0	Alarm; both transmitters of that inlet read bad ⇒ pair drops out	Reset XF05 (HMI, DO 0.0)

Fault	Detection	Automatic response	Recovery
Framo link / heartbeat lost	no clear for 10 s	Alarm *Communication loss*; PLC shows UNAVAILABLE; LIC01 output frozen if the pump was in level control; oxygenation permit off. PCC: HOLD at preset speed	Link back: PCC still in HOLD ⇒ operator Reset[OI2]; when REMOTE, LIC01 resumes from the actual speed
Pump trip / fault	PCC alarm flags; state LOCKDOWN	Common alarm *Pump n failed*; pump leaves level control; the remaining pumps carry the level (level dips, then LIC01 raises the speed)[OI6]; permit off	Fix the fault, operator Reset[OI3]; restart the pump
Start not acknowledged	Start sent, still AVAILABLE after 30 s	Alarm *Start failed*	Operator; [FD] wants the next pump started (HOLD)[OI15]
Stop not acknowledged	Stop sent, still in START, REMOTE or HOLD after 30 s	Alarm *Stop failed*	Operator, locally at the PCC
Speed deviation	$\lvert n_{SP} - n \rvert$ in REMOTE	not implemented[OI9]	–
Vacuum chamber overfill	LSH3n	not implemented (5 s delay in [FD])[OI11]	–
Emergency stop	PCC `EmergencyStop`	Pump stops (STO); alarm; LOCKDOWN	Release E-stop, Reset[OI13]
LIC01 at its limit	`LimitsActive`	Alarm; integral held (anti-windup of `Util.PID`)	Automatic

4.6 Interlocks

ID	Interlock	Source	In PLC
I1	Oxygenation permit line n = PM-3n in REMOTE AND heartbeat OK.	[FD] §3.2.1, [PID] -100	yes
I2	2oo3 high-high LS7nmHH in inlet box n stops PM-3n.	[FD] §3.2.6, [PID] -600	no[OI20]
I3	Group start only with the outlet open.	PLC function flow diagram	no[OI20]
I6	No valid level pair: LIC01 to manual at SafeOutput.	[PID] -400	yes
–	Start only when the pump is AVAILABLE and the link is OK.	[FD] Tabell 5	yes
–	Emergency stop: Framo E-stop stops the pump (STO); oxygen line to stop.	[FD] Tabell 2, [EL]	partly[OI13]

4.7 Communication

- **PLC ↔ Framo PCCn:** OPC UA. The PLC is the client (CODESYS Data Sources `Framo_1..3`); each PCC is a server (TwinCAT TF6100). Signal list: Table 13. Start, Stop and Reset are events acted on at the rising edge[OI4].
- **PLC ↔ InletRIO n:** Modbus TCP, function code 23, every 100 ms.
- **HMI ↔ PLC:** OPC UA (PLC is the server).

5 Non-functional requirements

5.1 Control law and tuning

LIC01 is a PI controller (CODESYS `Util.PID` wrapped in `PIDController`):

$$n_{\mathrm{SP}}(t) = K_p\Big(e(t) + \frac{1}{T_i}\int e\,dt\Big) + Y_{\mathrm{offset}}, \qquad e = \mathrm{SP} - \Delta H_{\mathrm{avg}},$$

limited to 40–130 rpm; the integral is held while the output is at a limit. Manual→auto resets the PID with $Y_{\mathrm{offset}} = Y_{\mathrm{prev}} - K_p e$ (bumpless).

Table 8: LIC01 start values.

Parameter	[PLC]	[FD] Tabell 8	Comment
K_p	450 rpm/m	P = 0.5 / 1 / 2 for 1 / 2 / 3 pumps	450 rpm/m ≈ [FD] P = 1 (90 rpm over 0.20 m). One set for all pump counts[OI5]
T_i	60 s	600 s	600 s gives a closed loop of about 16 min [DES]; 60 s tested[DEC4]
T_d	0	–	PI
Output	40–130 rpm	40–130 rpm	
PV scaling	m	0–20 cm	
Margins (worst case O-35, 3 pumps)	PM 76°, GM 13.2 dB, $1/\omega_c$ = 100 s	–	[DES] §7, dead time 2.5θ

Simulator tests (PFC200 against CasRioSim/CasFramoSim, O-35, 3 pumps, K_p 450, T_i 60 s):

- start-up: pumps reach REMOTE at about 87 rpm, take-over bumpless;
- setpoint step +2 cm: settles in about 100 s, no overshoot;
- one of three pumps lost: the level dips 3.4 cm; the pump rejoining gives 2.7 cm overshoot. There is no feed-forward on the number of pumps, and the integral winds up against 130 rpm while two pumps carry the load.

Proposal[OI6]: on a change of the pump count N, scale the output by $N_{\mathrm{old}}/N_{\mathrm{new}}$ (feed-forward, e.g. $N/(N-1)$ on a trip), with anti-windup reset to the scaled value, and schedule $K_p \propto 1/N$ (the process gain grows with N [DES]). An I-PD form would avoid the setpoint kick on mode changes.

Small setpoints. O-60 and S-75 use a 0.5 cm setpoint, against about 1 mm resolution of the over-height, ±6 mm combined transmitter uncertainty and waves up to $H_s = 2$ m. Whether this is controllable, and with what filter, is open[OI7].

5.2 Cycle times and tasks

A level reading reaches LIC01 two `Task` cycles after it is read; the speed setpoint reaches the PCC within some tenths of a second plus the OPC UA publishing time. This is small against the level loop's dead time of about 2 s and time constant of 28–40 s [DES]. Task watchdogs are disabled in the test application[OI23].

5.3 Signal resolution and filtering

- **750-455:** value in bits 14..4 (0..2047), bits 3..2 reserved, bits 1..0 diagnostics; 20 mA = `0x7FF0`[DEC2]. On the 0–2 m span one step is 0.98 mm; the over-height, the difference of two

Table 9: Tasks and timing ([PLC] Task Configuration of PFC200_Sim).

Task	Cycle	Inlet content
`Task`	20 ms	`Main`: InletSupervisor (LIC01, sequence), InletControl[0..2] (pump control, heartbeat, InletSensors), PenControl (PV); `IO_Processing` (InletRIO masters)
`OPC_Processing`	50 ms	`FramoDataProcessing`: maps Framo_n to/from GVLMain
`DatasourcesTask`	50 ms	OPC UA client (Data Sources) to the PCCs
Modbus TCP poll	100 ms	InletRIO n, FC23

readings, moves in steps of about 1 mm. The PLC scales Raw $\cdot$ (Max $-$ Min)/32760 + Min (20 mA reads 99.98 % of span).

- **Transmitter accuracy:** inside ±2 mm (radar), outside ±4 mm (pressure, 0.2 % of an assumed 2 m full scale)[OI18].
- **Filter:** 2nd-order Butterworth per pair, $f_c = 0.05$ Hz, bilinear with pre-warping, sample time = task cycle read from the runtime, LREAL coefficients. A 1 cm step reaches 63 % in 5.6 s with 4 % overshoot; 4–10 s waves are attenuated by about 19 dB [DES]. The filter restarts at the current value when a pair becomes valid.
- The per-transmitter filters (`FilterFc` 0.05 Hz) are for display only.
- **Signal direction:** raw 0 (4 mA) is assumed to be the lowest level; to be read in the transmitters. Inside and outside must agree, or the sign of LIC01 is wrong[OI18].

5.4 Alarm functions

Alarm states are handled by `AlarmManagement` (raised on the condition, acknowledged on the rising edge of the acknowledge bit). [AL] defines no priorities or acknowledgement classes; the column Class is a proposal (A = alarm, operator action; W = warning; N = notification only)[OI21].

Table 10: Alarms of the inlet system ([FD] Tabell 7 and 9, [AL], [PLC]). Per pump n = 1..3 unless noted.

Alarm	Condition	Class	Action	PLC
Communication loss pump n	PCC has not cleared HeartbeatToggle for 10 s	A	UNAVAILABLE; LIC01 frozen; O_2 line n stops	yes
Start of pump n failed	Start sent and still AVAILABLE after StartupTime (30 s)	A	none	yes
Stop of pump n failed	Stop sent and still in START, REMOTE or HOLD after ShutdownTime (30 s)	A	none	yes
Pump n failed (common)	any of DriveError, DriveCommunication, DriveNotInRemote, MotorTemperature, EmergencyStop, DCUndervoltageError, PumpStopped, UPSSupplyError, EmergencyRunMode	A	shown on info panel	yes[OI10]

Alarm	Condition	Class	Action	PLC
Individual Framo alarms	each flag above, own alarm	A	logged	yes
Pump n low / high power	LowPower / HighPower from PCC	N	none, notification only [FD]	wrongly also in common OI10
Speed deviation pump n	$\lvert n_{\mathrm{SP}} - n \rvert >$ TBD rpm for TBD s, in REMOTE	W	none	no OI9
Vacuum overfill line n	LSH3n for 5 s (configurable)	A	none	no OI11
Emergency stop activated	EmergencyStop from PCC (hard-wired TBD)	A	pump stops	via Framo OI13
Level control deviation	$\lvert \mathrm{PV} - \mathrm{SP} \rvert >$ TBD for TBD s, LIC01 in auto	W	none	no OI8
Low level in pen	$\Delta H_{\mathrm{avg}} <$ TBD for TBD s	A	none	no OI8
High level in pen	$\Delta H_{\mathrm{avg}} >$ TBD for TBD s	A	none	no OI8
Level transmitter fault (LT01–03, LT21–23)	750-455 status bits $\neq$ 00 for 1 s	W	pair drops out of PV	generic quality alarm OI8
Level measurement lost (LAL01)	no valid pair (pen)	A	LIC01 at SafeOutput	yes
LIC01 output at limit / overflow	`LimitsActive`, `Overflow` (pen)	W	none	yes
InletRIO n communication error	Modbus `CommOK` FALSE	A	see OI17	yes
InletRIO n field power	XF05 tripped	A	LTs of inlet n lost	yes

5.5 Parameters

All parameters are written from the HMI and retained: `Main.UpdateRetain` keeps them in step with `GVLRetain`. Defaults from `ConfigValues` apply on a fresh download.

Table 11: Configurable parameters of the inlet system.

Parameter	Default	Unit	Meaning
`PenConfig.LevelController.Kp`	450	rpm/m	LIC01 gain
`PenConfig.LevelController.Ti`	60	s	integral time (new value takes effect at the next start or manual→auto)
`PenConfig.LevelController.Td`	0	s	derivative time
`PenConfig.LevelController.MinOutput` / `MaxOutput`	40 / 130	rpm	output limits
`PenConfig.LevelController.SafeOutput`	60	rpm	speed with no valid level pair
`Pen.Command.LevelSetpoint`	0.06	m	over-height setpoint (O-35)
`LevelInsideConfig[i]`: Enable, MinValue, MaxValue	TRUE, 0, 2.0	m	LT0i in use and range
`LevelInsideConfig[i]`: FilterFc, QualityEnabled	0.05, TRUE	Hz	display filter; quality alarm
`LevelOutsideConfig[i]`: as inside	as inside		LT2i

Parameter	Default	Unit	Meaning
`OverHeightConfig[i]`: Enable, FilterFc	TRUE, 0.05	Hz	pair in use; cut-off of the pair's filter (0 = off)
`InletConfig[i].HeartbeatInterval`	1	s	PLC sets the bit this long after the PCC cleared it
`InletConfig[i].HeartbeatTimeout`	10 (DUT 5)	s	link-loss detection
`InletConfig[i].StartupTime` / `ShutdownTime`	30 / 30	s	start / stop failed alarms
`InletRIOConfig[i]`: IP, Port	sim: 10.1.2.101, 5121–5123		coupler address (site plan TBD)
Data sources `Framo_1..3`	per project		OPC UA endpoint of each PCC OI8 OI9 OI11 OI14
Not yet defined: deviation limit and delay, L/H level limits and delays, speed deviation limit, vacuum overfill delay (5 s), mode table SP	TBD		

5.6 Interfaces: instruments and actuators

Table 12: Instruments, actuators and PLC functions of the inlet system ([PID], [EL], [DES]). Use: C = control, M = monitor, A = alarm.

Tag	Function	Location	Signal / interface	Range	Resolution	Use
LT01–03	Level inside the pen, at inlet n; VEGAPULS radar OI18	Gooseneck, inlet n	4–20 mA 2-wire, InletRIO n -R1-106 ch1, word 0	0–2 m	0.98 mm; ±2 mm	C, M, A
LT21–23	Sea level outside the pen, at inlet n; submersible pressure probe	Outside the collar, inlet n	4–20 mA 2-wire, InletRIO n ch2, word 1	0–2 m (assumed)	0.98 mm; ±4 mm	C, M, A
LDY01–03	Over-height of pair n, filtered	PLC	computed	0–0.20 m	≈1 mm	C
LY01	Mean over-height of valid pairs (PV)	PLC	computed	0–0.20 m	–	C, A
LIC01	Level controller, PI	PLC	output to SY31–33	40–130 rpm	–	C
LAL01	Level measurement lost	PLC	alarm	–	–	A
PM-31–33	Framo SR2000 inlet pump, rim motor 60 kW	Intake n	via PCCn (ACS880)	40–130 rpm, 20 000 m^3/h	–	actuator
SC31–33	Framo pump controller CX8200 + ACS880-31-105A-3	PCCn	OPC UA server	–	–	–
SY31–33	Speed setpoint to pump n	PLC → PCCn	OPC UA SpeedSetpoint	40–130 rpm	REAL	C
SI31–33	Actual pump speed	PCCn	OPC UA MotorVelocity	0–130 rpm	REAL	C, M
HS31–33	Start / Stop / Reset pump n	PLC → PCCn	OPC UA events	–	–	C
YI31–33	Pump state	PCCn	OPC UA 7 flags	7 states	–	C, M
UA31–33	Pump alarms (11 flags)	PCCn	OPC UA alarms	–	–	A

Tag	Function	Location	Signal / interface	Range	Resolution	Use
KY31–33	Heartbeat	PLC ↔ PCC*n*	OPC UA BOOL	–	–	C, A
LSL31–33	Vacuum chamber low level (air evacuated)	TV-3*n*	OPC UA (Framo LSL01)	on/off	–	M
LSH31–33	Vacuum chamber high level (overfill)	TV-3*n*	OPC UA (Framo LSH02)	on/off	–	A[OI11]
PT31–33	Compressed air pressure to ejector	Air line, inlet *n*	OPC UA analog	Framo range	REAL	M
XV31A–33A	Air valve to ejector EJ-3*n*	Inlet *n*	status OPC UA (V001?)	open/closed	–	M[OI11]
XV31B–33B	Vacuum break (vent) valve	TV-3*n*	status OPC UA (V002?)	open/closed	–	M[OI11]
E-stop PCC*n*	Local emergency stop, STO	PCC*n* door	OPC UA EmergencyStop	on/off	–	A[OI13]
YA30 (n)	24 V field breaker XF05 tripped (LT supply)	RIO *n* cabinet	InletRIO *n* DI word 4.0	on/off	–	A

5.7 Interfaces: OPC UA signals to and from the Framo controllers

Per pump *n* the PLC uses the Data Source `Framo_n`; the GVL side is `GVLMain.Inlet[n-1].InletPump`. Direction is seen from the PLC (R = read, W = written).

Table 13: OPC UA signal list, PLC ↔ Framo PCC*n* ([FRAMO] rev. B, [FD] Tabell 6, [PLC] FramoDataProcessing).

Signal (Data Source name)	Type	Dir.	Use in PLC	Remark
MotorVelocity (+ ValueRange Low/High)	AnalogValue	R	tracking, bumpless transfer, speed indication	range written by Framo?[OI19]
MotorFrequency, MotorCurrent, MotorPower, MotorVoltage, MotorTorque, PowerConsumptionCounter, DCLinkVoltage, VSDHeatSinkTemperature, VSDOperatingDays	AnalogValue	R	info panel	[FRAMO]: PowerConsumtionCounter
MotorRunningDays	AnalogValue	R	info panel; run-time order of pumps (`SortSequence`)	order not yet used[OI15]
DriveError, DriveCommunication, DriveNotInRemote, MotorTemperature, EmergencyStop, DCUnderVoltageError, PumpStopped, UPSSupplyError, EmergencyRunMode	Alarm	R	own alarm each; common alarm	
LowPower, HighPower	Alarm	R	notification	[OI10]
PumpStateUnavailable, ... Available, Start, Stop, Remote, Hold, Lockdown	Event	R	`InletPump.State` (first TRUE flag)	

Signal (Data Source name)	Type	Dir.	Use in PLC	Remark
Start, Stop, Reset	Event	W	commands, rising edge	OI3OI4
SpeedSetpoint	AnalogValue	W	LIC01 output in REMOTE, else actual speed or 0	[FRAMO]/[FD]: VelocitySetpoint[OI19]
HeartbeatToggle	BOOL	R/W	PLC sets, PCC clears	OI1
HeartbeatInterval, HeartbeatElapsed	TIME	–	not mapped	OI1
PressureCompressedAir, LevelSwitchLSL01, LevelSwitchLSH02, StatusV001, StatusV002	Analog / BOOL	R	vacuum system status	not in [FRAMO] / [FIL][OI11]

5.8 HMI operator functions

Table 14: Operator functions for the inlet system (PLC variables in [PDS] Table “Commands”).

Function	PLC variable	Note
Pen Start / Pen Stop	`Pen.Command.Start`, `.Stop`	pulse; starts/stops all pumps in auto
Level setpoint	`Pen.Command.LevelSetpoint`	m; from the mode in future[OI14]
LIC01 auto / manual, manual output	`LevelController.Command.Manual`, `.Auto`, `.ManualOutput`	rpm
Pump auto / manual	`Inlet[i].InletPump.Command.Auto`, `.Manual`	default auto
Pump start / stop / speed in manual	`Inlet[i].InletPump.Command.Start`, `.Stop`, `.Setpoint`	speed used in REMOTE
Reset pump (HOLD, LOCKDOWN)	`Inlet[i].InletPump.Command.Reset`	sent to the PCC
Enable/disable transmitter or pair	`...LevelInsidePen.Config.Enable`, `...PenOverHeight.Config.Enable`	takes a pair out of the PV
Info panel per pump	`Inlet[i].InletPump.Input.*`, `.State`	[FD] Figur 3
Alarm list and acknowledgement	`*.Alarm.*`, `*.AlarmAck.*`	classes TBD[OI21]
Parameters (Table 11)	`GVLMain.PenConfig`, `InletConfig`, ...	access level TBD

6 Open issues and decisions

6.1 Open issues

Each item names the sources that disagree and, where there is one, a proposal. “Framo” items need agreement with the supplier.

OI1 Heartbeat baseline. PLC: one shared BOOL, PLC sets / PCC clears, neither side waits after connect, interval 1 s, timeout 10 s (DUT default 5 s). [FRAMO] rev. B: HeartbeatToggle with HeartbeatInterval and HeartbeatElapsed. [FD]: INT counter 0–32767. Agree one with Framo, incl. the PCC’s own timeout and whether the PLC writes Interval.

OI2 HOLD → REMOTE. [FD] and the PLC require an operator Reset; [FRAMO] rev. B returns to REMOTE automatically when the heartbeat is back (and so does the simulator).

OI3 LOCKDOWN exit and Reset. [FD]: operator Reset after the fault is fixed. [FRAMO]: LOCKDOWN ends with "sequence end" to UNAVAILABLE, no Reset. What Reset does (all alarms? edge?) is not described (Framo review comment).

OI4 Stop and command semantics. [FRAMO] rev. B accepts Stop only in REMOTE (not in START or HOLD). The PLC sends Stop in any state and raises Stop failed for START, REMOTE and HOLD. Are Start/Stop/Reset edge- or level-triggered?

OI5 LIC01 parameter sets. [FD]: three sets by number of pumps (P 0.5/1/2, I 600 s); PLC: one set (K_p 450 rpm/m, T_i 60 s). [DES]: the gain should fall with more pumps, not rise. Decide.

OI6 Pump-count transients. Losing 1 of 3 pumps dips the level 3.4 cm; rejoining overshoots 2.7 cm. Proposal: $N_{\text{old}}/N_{\text{new}}$ feed-forward, anti-windup reset, gain scheduling $K_p \propto 1/N$; consider I-PD against setpoint kick.

OI7 0.5 cm setpoint (O-60, S-75). Controllable with ≈1 mm resolution, ±6 mm uncertainty and waves? Decide the setpoint, filter and acceptance criterion.

OI8 Pen level alarms. Deviation, low and high level alarms of [FD] Tabell 9 / [AL] are not in the PLC; limits and delays TBD. The per-transmitter fault exists only as the generic quality alarm; map it to the [AL] text per tag.

OI9 Speed deviation alarm (in REMOTE) of [FD] Tabell 7 not implemented; limit and delay TBD.

OI10 LowPower / HighPower are included in `PumpCommonError`; [FD]: notification only. Remove them from the common alarm.

OI11 Vacuum system. Alarms not implemented (`VacuumSystemAlarmDUT` is empty; LSH overfill with 5 s delay). The vacuum signals are not in [FRAMO] or [FIL]. Tag mapping to confirm: Framo LSL01/LSH02/V001/V002 vs [PID] LSL3n/LSH3n/XV3nA/B vs [FD] LSL31/LSH32/XV31/XV32.

OI12 Flow figures. [FD] Tabell 10 O-35 = 34 000 m^3/h; [DES] maximum 27 000 m^3/h; 3 pumps × 20 000 m^3/h rated. Framo comment: "26 000 ok for emergency mode?". Align.

OI13 Emergency stop. Do the Framo E-stops (in series, [EL] 874.001) stop all three lines? How is oxygenation stopped on E-stop through the Heimdall PLC (no E-stop input wired today)? [FD] alarm "Nødstopp aktivert" is HOLD.

OI14 Setpoint from operating mode. [FD]: LIC01 setpoint chosen from the mode, with manual override. PLC: operator-entered setpoint only; the outlet mode table has damper positions only.

OI15 Number of pumps and rotation. [FD] (HOLD): stop a pump when LIC01 stays at minimum, start one when it stays high, start the next if a start fails, warn when a pump has been idle too long, rotate by run time. PLC: `SortSequence` orders by run time but is not used; `NumberOfInletsToRun` unused.

OI16 Auto-restart and start order. [FD] Tabell 2–3: restart after power failure in a sequence (2a/2b/2c). PLC: pen start latch not retained; Pen Start sends Start to all pumps at once.

OI17 InletRIO link loss. The last level words are kept and the pair stays valid, so LIC01 controls on a frozen PV. Proposal: treat a pair as invalid while its RIO has no communication.

OI18 **Level transmitters.** Signal direction (4 mA = lowest level?) to be read in the sensors; inside model VEGAPULS C21 ([DES]) vs C22 ([PID] loop sheet); outside type and 2 m full scale assumed.

OI19 **Framo interface details.** Names differ (SpeedSetpoint vs VelocitySetpoint, DCUnder-VoltageError, PowerConsumptionCounter); is the MotorVelocity range written by Framo? Hardware CX8190 ([FRAMO]) vs CX8200 ([EL]); pump SX1000 ([EL]) vs SR2000 ([FD]).

OI20 **Interlocks from other systems.** 2oo3 LS7nmHH trip of PM-3n (I2) and "group start only with the outlet open" (I3) are not in the PLC.

OI21 **Alarm classes.** [AL] has no priority or acknowledgement class; define them.

OI22 **Framo dynamics and preset speeds.** Speed response T_p (10 s assumed), START preset speed (88 rpm assumed), HOLD safe speed and ramp rates: to be given by Framo.

OI23 **Platform.** Task watchdogs disabled in the test application; IP plan and network drawing missing ([PID] C3).

6.2 Decisions

DEC1 2026-10-15 (PLC, in review): PV = mean of the per-pair filtered ΔH_i over the valid pairs; a lost transmitter drops its pair ([FD] §3.2.2).

DEC2 2026-10-07: 750-455 word layout confirmed by WAGO (value in bits 14..4, 0..2047).

DEC3 2026-10-15 (PLC, in review): no valid pair ⇒ alarm LevelMeasurementLost and LIC01 in manual at SafeOutput (60 rpm), bumpless return.

DEC4 2026-10-13: LIC01 start value T_i = 60 s instead of 600 s, from the design basis and the simulator test; to be confirmed at commissioning.